Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/11/2017
PI Name:	Bailey, Michael R. Ph.D.		
Project Title:	Prevention of Renal Stone Complications in Space Exploration		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	(1) **Medical Conditions**:Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures (2) **Renal Stone**:Risk of Renal Stone Formation		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	bailey@apl.washington.edu	**Fax:**	FY 206-221-6578
PI Organization Type:	UNIVERSITY	**Phone:**	206-685-8618
Organization Name:	University of Washington		
PI Address 1:	Applied Physics Laboratory/CIMU		
PI Address 2:	1013 NE 40th Street		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98105-6698	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Crew Health NNJ12ZSA002N
Start Date:	06/01/2013	**End Date:**	12/31/2016
No. of Post Docs:	12	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2016 per NSBRI (Ed., 3/11/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wang, Yak-Nam (University of Washington) Sorensen, Mathew (University of Washington) Khohklova, Vera (M.V.Lomonosov Moscow State University) Sapozhnikov, Oleg (University of Washington) Crum, Lawrence (University of Washington) Harper, Jonathan David (University of Washington) Kreider, Wayne (University of Washington)		
Grant/Contract No.:	NCC 9-58-SMST03402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	1. Specific aims We will refine and validate probes to integrate with the NASA Flexible Ultrasound System (FUS) to address Exploration Medical Capabilities (ExMC) Gap 4.02 Nephrolithiasis. AIM 1. Refine ultrasound probes to detect, reposition, and fragment kidney stones. AIM 2. Validate probes to visualize, reposition, and fragment stones. AIM 3. Refine and validate imaging to guide therapy. 2. Key Findings A probe and software to image and reposition kidney stones were developed and integrated on a radiation hardened flexible ultrasound system (FUS) and demonstrated effectively on human subjects. A probe to image, reposition, and fragment stones was designed, fabricated, and integrated into an FUS and is currently in clinical trials to expel stone fragments. Software was developed and integrated on an FUS and validated in human subjects to improve kidney stone detection and size determination. The ability to reposition stones was also integrated into the partially completed NASA FUS with the NASA FUS probes and demonstrated. The work has garnered attention. Reports have been sent to NSBRI (National Space Biomedical Research Institute), FDA (Food & Drug Administration), NIH (National Institutes of Health), NASA, and OMB (Office of Management and Budget). Demonstrations have been conducted at American Urological Association (AUA) annual meetings each year, Congress twice, and several other professional societies. Over 40 papers have been published. Over 40 patent applications have been submitted. Students, residents, and fellows have trained on the project. Technology developed in this research has been licensed to a spin-off company SonoMotion Inc. 3. Impact We have invented a technology to reposition kidney stones and demonstrated it works in people. In four of the cases, what appeared as one large stone on x-ray was two or three small passable stones. This had direct diagnostic benefit to these subjects and changed their course of treatment. In four other subjects, we moved stones out of the kidney, which they passed. This result was a direct therapeutic benefit to these subjects. One subject felt relief from a painful obstructing stone. We have shown we can produce a working prototype, develop sufficiently high-quality imaging to guide treatment, train new users, and conduct a successful clinical trial. We refined the system design, submitted for publication in vitro results quantifying the improvement, and entered a second clinical trial. The refined design also has the capability to fragment stones. This design is being commercialized. Specifically, we have now implemented our technologies with different probes making it efficient to add the probes NASA selects or to continue to refine the probes we can provide. Our imaging software can be added to an FUS or commercial imager. Our pushing capability has been added as a software upgrade to the FUS. Our advanced repositioning and fragmenting probe is readily integrated with any standard or FUS imager with minimal additional mass and software change to the system. Our final system and the system being commercialized, when validated in human in a flight analog, largely close the gap of nephrolithiasis or exploration mission and extends application to the emergency department on Earth. Our new stone sizing technique can be used on any imager by any user to improve the accuracy of stone size determination with ultrasound. Overestimated stone size leads to unnecessary surgeries, and underestimated stone size leads to obstructions and ER (emergency room) visits. Stone size similarly determines risk and course of action in space. 4. Proposed Research We are conducting a clinical trial of S-mode software for automatic stone detection and stone sizing. We are conducting a clinical trial of expelling stone fragments. We have received approval and set up the infrastructure for an test of ultrasonic propulsion in an Emergency Department (ED) analog to a space emergency, and seek funding for that trial. We are testing safety and effectiveness in clinical simulation in animal studies of stone breaking to add this capability to our ED trial. The technology is also being tested for gallstones.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Kidney stones have long been near the top of NASA's list of concerns; mitigating Gap 4.02 medical condition Nephrolithiasis is a shall for all missions beyond the International Space Station (ISS). Likewise, stones have plagued humans since ancient Egypt. Currently, one in eleven Americans has suffered from stones -- more than have diabetes or cardiovascular disease. Dehydration, stasis, and bone demineralization are strong contributors to kidney stones, and occur in microgravity, increasing the risk of stones in space. Stones are often debilitating, and pilots cannot fly with stones. Stones occurred on a Russian space mission, and the mission was nearly aborted before the stone passed. Over 30 stones have occurred shortly following even short duration space flights. NASA has collected compelling evidence for concern on its website https:// . Additionally, since the website publication, the total number of astronaut stone episodes has more than doubled, and a drug introduced to combat visual impairment/intracranial pressure has exacerbated the risk. Science, experience, and the negative medical consequences support concern for the risk of stones in space. NASA and NSBRI have focused considerable attention on stones and made progress. However, there are many types of stone disease, and it is unlikely that stone disease will ever be completely prevented on Earth or in space. We propose a way to prevent or minimize the consequences of any stones that form while in space. The treatment for most kidney stones is to encourage natural passage. To quote NASA's expectations in space Based on current Lifetime Surveillance of Astronaut Health (LSAH) data, 80 to 85% of in-flight cases of nephrolithiasis are expected to be best case scenarios (defined as a renal stone that responds to conservative treatment, e.g., analgesics and hydration), and 15 to 20% would be worst case scenarios (defined as a renal stone that does not respond to conservative treatment, e.g., requires lithotripsy or surgical treatment). Even surgery leaves residual fragments that must pass. Our technology provides the capability to reposition stones within the kidney and ureter, which will enhance conservative treatment or surgery by accelerating and facilitating passage of stones or fragments. However, this does not have to be the only use. The technology can also be used to reposition a stone to a non-obstructing location within the kidney to postpone surgery or to accelerate passage through the ureter, as proposed here. Finally, the technology proposed in this grant also provides the capability to comminute the stone as in shock wave lithotripsy (SWL) with what we call burst wave lithotripsy (BWL).

Task Progress:	All tasks were completed. AIM 1. Refine the ultrasound probes to detect, reposition, and fragment kidney stones. Task 1.1. Select imaging probe for stone repositioning. We enabled the capability to push stones on the as yet incomplete NASA FUS system with the NASA GE C1-6 abdominal imaging probe. These results were reported in an NSBRI Advanced Technology Demonstration: Prevention of Renal Stone Complications in Space Exploration in August 2016. Deliverable of the grant The capability to reposition kidney stones noninvasively has been added to the NASA flexible ultrasound system. Task 1.2. Custom design probe to image, reposition, and fragment stones. We invented Burst Wave Lithotripsy which has fully comminuted stones of all compositions in under 20 minutes and has fragmented many stones in seconds. All work so far has been in water tanks or animals, not humans. The size fragments are controlled by the ultrasound frequency. The technique reduces peak pressure by 1/10th but increases pulse duration about five times and pulse repletion rate at least 20 times to deliver more energy more quickly and possibly without discomfort. AIM 2. Validate probes to visualize, reposition, and fragment kidney stones. Task 2.1. Validate capability to displace an obstructing stone. Task 2.2. Validate capability to displace a ureter stone. Task 2.3. Validate capability to comminute a stone. These tasks were completed. A small probe embedded centrally within the Burst Wave Lithotripsy (BWL) therapy probe connected to the Verasonics FUS system provides image guidance for BWL. We have targeted and fragmented human stones surgically placed in over 10 pigs. Preclinical test results have been submitted for publication. A clinical trial of the system is underway. AIM 3. Refine and validate imaging to guide therapy. Task 3.1. Refine and validate capability to measure the size of kidney stones. In a series of 3 papers, we demonstrated how ultrasound imaging can be optimized to accuracy similar to CT with user controls as well as software modifications. The imaging we have implemented on the Verasonics FUS appears to size stones more accurately than clinical imagers. Task 3.2. Refine capability to localize a stone. Task 3.3. Refine and validate capability to detect a ureter stone. We have developed enhanced B-mode, enhanced Doppler-based twinkling, and combined them into a stone specific imaging mode called S-mode. S-mode has been published. S-mode has been used to image stones in the kidneys and ureters in human subjects. In our most recent study of hundreds of imaging frames from 40 stones and 28 subjects, the signal to noise ratio of S-mode was ten times the grayscale SNR. This paper won Best Abstract at the Engineering and Urology meeting of the AUA in 2016.
Bibliography Type:	Description: (Last Updated: 07/29/2025)
Abstracts for Journals and Proceedings	Janssen KM, Brand TC, Bailey MR, Cunitz BW, Sorensen MD, Harper JD, Dunmire B. "Effect of stone size and composition on ultrasonic propulsion in vitro." 2017 American Urological Association Annual Meeting, Boston, MA, May 12-16, 2017. Journal of Urology. 2017 Apr;197(4 Suppl):e835. (2017 Annual Meeting Program Abstracts, AUA Annual Meeting, Boston, MA, May 12-16, 2017.) https://doi.org/10.1016/j.juro.2017.02.1951 , Apr-2017
Articles in Other Journals or Periodicals	Sorensen MD, Bailey MR. "Shock Wave Lithotripsy: Application and future direction." Scientific American. 2017 in press as of May 2017. , May-2017
Articles in Peer-reviewed Journals	Karzova MM, Yuldashev PV, Sapozhnikov OA, Khokhlova VA, Cunitz BW, Kreider W, Bailey MR. "Shock formation and nonlinear saturation effects in ultrasound field of a diagnostic curvilinear probe." J Acoust Soc Am. 2017 Apr;141(4):2327. http://dx.doi.org/10.1121/1.4979261 ; PubMed PMID: 28464662 , Apr-2017
Articles in Peer-reviewed Journals	Bailey MR. "Evaluation of renal calculi passage while riding a roller coaster. (Comment on Mitchell and Wartinger article, 'Validation of a functional pyelocalyceal renal model for the evaluation of renal calculi passage while riding a roller coaster.' PubMed PMID: 27669068)" Journal of the American Osteopathic Association. 2017 Jun 1;117(6):349-50. https://doi.org/10.7556/jaoa.2017.069 ; PubMed PMID: 28556853 , Jun-2017
Articles in Peer-reviewed Journals	Rosnitskiy PB, Yuldashev PV, Sapozhnikov OA, Maxwell AD, Kreider W, Bailey MR, Khokhlova VA. "Design of HIFU transducers for generating specified nonlinear ultrasound fields." IEEE Trans Ultrason Ferroelectr Freq Control. 2017 Feb;64(2):374-90. http://dx.doi.org/10.1109/TUFFC.2016.2619913 ; PubMed PMID: 27775904; PubMed Central PMCID: PMC5300962 , Feb-2017
Articles in Peer-reviewed Journals	May PC, Bailey MR, Harper JD. "Ultrasonic propulsion of kidney stones. (Technological Innovations in Urologic Surgery section)" Curr Opin Urol. 2016 May;26(3):264-70. Review. http://dx.doi.org/10.1097/MOU.0000000000000276 ; PubMed PMID: 26845428; PubMed Central PMCID: PMC4821680 , May-2016
Articles in Peer-reviewed Journals	May PC, Kreider W, Maxwell AD, Wang YN, Cunitz BW, Blomgren PM, Johnson C, Park JSH, Bailey MR, Lee D, Harper JD, Sorensen MD. "Detection and evaluation of renal injury in burst wave lithotripsy using ultrasound and magnetic resonance imaging." Journal of Endourology. 2017 Aug;31(8):786-92. Epub 2017 May 18. http://dx.doi.org/10.1089/end.2017.0202 ; PubMed PMID: 28521550 , Aug-2017

Articles in Peer-reviewed Journals	May PC, Haider Y, Dunmire B, Cunitz BW, Thiel J, Liu Z, Bruce M, Bailey MR, Sorensen MD, Harper JD. "Stone-mode ultrasound for determining renal stone size." J Endourol. 2016 Sep;30(9):958-62. http://dx.doi.org/10.1089/end.2016.0341 ; PubMed PMID: 27393000; PubMed Central PMCID: PMC5031098 , Sep-2016
Articles in Peer-reviewed Journals	Simon JC, Dunmire B, Bailey MR, Sorensen MD. "Developing complete ultrasonic management of kidney stones for spaceflight." J Space Safety Eng. 2016 Sep;3(2):50-7. Review. http://dx.doi.org/10.1016/S2468-8967(16)30018-0 , Sep-2016
Articles in Peer-reviewed Journals	Khokhlova VA, Yuldashev PV, Rosnitskiy PB, Maxwell AD, Kreider W, Bailey MR, Sapozhnikov OA. "Design of HIFU transducers to generate specific nonlinear ultrasound fields." Physics Procedia. 2016;87:132-8. (45th Annual Symposium of the Ultrasonic Industry Association, UIA 45th Symposium, 4-6 April 2016, Seattle, WA, USA) https://doi.org/10.1016/j.phpro.2016.12.020 , Dec-2016
Articles in Peer-reviewed Journals	Brisbane W, Bailey MR, Sorensen MD. "An overview of kidney stone imaging techniques." Nat Rev Urol. 2016 Nov;13(11):654-62. Epub 2016 Aug 31. Review. http://dx.doi.org/10.1038/nrurol.2016.154 ; PubMed PMID: 27578040 , Nov-2016
Articles in Peer-reviewed Journals	Oweis GF, Dunmire BL, Cunitz BW, Bailey MR. "Non-invasive measurement of the temperature rise in tissue surrounding a kidney stone subjected to ultrasonic propulsion." Conf Proc IEEE Eng Med Biol Soc. 2015 Aug;2015:2576-9. http://dx.doi.org/10.1109/EMBC.2015.7318918 ; PubMed PMID: 26736818; PubMed Central PMCID: PMC4832570 , Aug-2015
Articles in Peer-reviewed Journals	Khokhlova TD, Haider YA, Maxwell AD, Kreider W, Bailey MR, Khokhlova VA. "Dependence of boiling histotripsy treatment efficiency on HIFU frequency and focal pressure levels." Ultrasound Med Biol. 2017 Sep;43(9):1975-85. Epub 2017 Jun 20. https://doi.org/10.1016/j.ultrasmedbio.2017.04.030 ; PubMed PMID: 28641910 , Sep-2017
Articles in Peer-reviewed Journals	Cunitz BW, Dunmire B, Bailey MR. "Characterizing the acoustic output of an ultrasonic propulsion device for urinary stones." IEEE Trans Ultrason Ferroelectr Freq Control. 2017 Dec;64(12):1818-27. Epub 2017 Oct 2. https://doi.org/10.1109/TUFFC.2017.2758647 ; PubMed PMID: 28981413; PubMed Central PMCID: PMC5733808 , Dec-2017
Articles in Peer-reviewed Journals	Janssen KM, Brand TC, Bailey MR, Cunitz BW, Harper JD, Sorensen MD, Dunmire B. "Effect of stone size and composition on ultrasonic propulsion ex vivo." Urology. 2018 Jan;111:225-9. Epub 2017 Sep 28. https://doi.org/10.1016/j.urology.2017.09.013 ; PubMed PMID: 28964820 , Jan-2018
Articles in Peer-reviewed Journals	Simon JC, Sapozhnikov OA, Kreider W, Breshock M, Williams JC, Bailey MR. "The role of trapped bubbles in kidney stone detection with the color Doppler ultrasound twinkling artifact." Phys Med Biol. 2018 Jan 9;63(2):025011. PubMed PMID: 29131810; PubMed Central PMCID: PMC5791757 , Jan-2018
Articles in Peer-reviewed Journals	Dai JC, Dunmire B, Sternberg KM, Liu Z, Larson T, Thiel J, Chang HC, Harper JD, Bailey MR, Sorensen MD. "Retrospective comparison of measured stone size and posterior acoustic shadow width in clinical ultrasound images." World J Urol. 2018 May;36(5):727-32. Epub 2017 Dec 14. https://doi.org/10.1007/s00345-017-2156-8 ; PubMed PMID: 29243111 [reported originally in May 2017 as Epub ahead of print] , May-2018
Articles in Peer-reviewed Journals	Ghanem MA, Maxwell AD, Kreider W, Cunitz BW, Khokhlova VA, Sapozhnikov OA, Bailey MR. "Field characterization and compensation of vibrational non-uniformity for a 256-element focused ultrasound phased array." IEEE Trans Ultrason Ferroelectr Freq Control. 2018 Sep;65(9):1618-30. Epub 2018 Jun 27. https://doi.org/10.1109/TUFFC.2018.2851188 ; PubMed PMID: 29994675 , Sep-2018
Articles in Peer-reviewed Journals	Maxwell AD, Wang YN, Kreider W, Cunitz BW, Starr F, Lee D, Nazari Y, Williams JC Jr, Bailey MR, Sorensen MD. "Evaluation of renal stone comminution and injury by burst wave lithotripsy in a pig model." J Endourol. 2019 Oct;33(10):787-92. https://doi.org/10.1089/end.2018.0886 ; PubMed PMID: 31016998; PMCID: PMC6798804 [note originally reported in Sept 2019 as "Published Online:27 May 2019."] , Oct-2019
Articles in Peer-reviewed Journals	Simon JC, Holm JR, Thiel J, Dunmire B, Cunitz BW, Bailey MR. "Evidence of microbubbles on kidney stones in humans." Ultrasound Med Biol. 2020 Jul;46(7):1802-7. Epub 2020 Apr 1. https://doi.org/10.1016/j.ultrasmedbio.2020.02.010 ; PMID: 32245546; PMCID: PMC7293935 , Jul-2020
Articles in Peer-reviewed Journals	Hall MK, Thiel J, Dunmire B, Samson PC, Kessler R, Sunaryo P, Sweet RM, Metzler IS, Chang HC, Gunn M, Dighe M, Anderson L, Popchoi C, Managuli R, Cunitz BW, Burke BH, Ding L, Gutierrez B, Liu Z, Sorensen MD, Wessells H, Bailey MR, Harper JD. "First series using ultrasonic propulsion and burst wave lithotripsy to treat ureteral stones." J Urol. 2022 Nov 1;208(5):1075-82. https://pubmed.ncbi.nlm.nih.gov/36205340 ; PMID: 36205340; PMCID: PMC10089227 , Nov-2022
Articles in Peer-reviewed Journals	Harper JD, Lingeman JE, Sweet RM, Metzler IS, Sunaryo P, Williams JC, Jr., Maxwell AD, Thiel J, Cunitz BW, Dunmire B, Bailey MR, Sorensen MD. "Fragmentation of stones by burst wave lithotripsy in the first 19 humans." J Urol. 2022 May;207(5):1067-76. http://dx.doi.org/10.1097/JU.0000000000002446 ; PMID: 35311351; PMCID: PMC9078634 , May-2022
Articles in Peer-reviewed Journals	Maxwell AD, Yuldashev PV, Kreider W, Khokhlova TD, Schade GR, Hall TL, Sapozhnikov OA, Bailey MR, Khokhlova VA. "A prototype therapy system for transcutaneous application of boiling histotripsy." IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control. 2017 Oct;64(10):1542-1557. https://doi.org/10.1109/TUFFC.2017.2739649 ; PubMed PMID: 28809681 [reported originally in May 2017 as in press] , Oct-2017
Articles in Peer-reviewed Journals	Dunmire B, Harper JD, Cunitz BW, Lee FC, Hsi R, Liu Z, Bailey MR, Sorensen MD. "Use of the acoustic shadow width to determine kidney stone size with ultrasound." J Urol. 2016 Jan;195(1):171-7. Epub 2015 Aug 22. http://dx.doi.org/10.1016/j.juro.2015.05.111 ; PubMed PMID: 26301788; PubMed Central PMCID: PMC4821497 , Jan-2016

Articles in Peer-reviewed Journals	Cunitz BW, Harper JD, Sorensen MD, Haider Y, Thiel J, May PC, Liu Z, Bailey MR, Dunmire B, Bruce M. "Quantification of renal stone contrast with ultrasound in human subjects." J Endourol. 2017 Nov;31(11):1123-30. https://doi.org/10.1089/end.2017.0404 ; PubMed PMID: 28847171; PubMed Central PMCID: PMC5695736 [reported originally in May 2017 as "in review"] , Nov-2017
Articles in Peer-reviewed Journals	Lee FC, Hsi RS, Sorensen MD, Paun M, Dunmire B, Liu Z, Bailey M, Harper JD. "Renal vasoconstriction occurs early during shockwave lithotripsy in humans." J Endourol. 2015 Dec;29(12):1392-5. Epub Aug 2015. http://dx.doi.org/10.1089/end.2015.0315 ; PubMed PMID: 26239232; PubMed Central PMCID: PMC4677566 , Dec-2015
Articles in Peer-reviewed Journals	Simon JC, Wang YN, Cunitz BW, Thiel J, Starr F, Liu Z, Bailey MR. "Effect of carbon dioxide on the twinkling artifact in ultrasound imaging of kidney stones: A pilot study." Ultrasound Med Biol. 2017 May;43(5):877-83. Epub 2017 Feb 9. http://dx.doi.org/10.1016/j.ultrasmedbio.2016.12.010 ; PubMed PMID: 28190622; PubMed Central PMCID: PMC5385287 , May-2017
Articles in Peer-reviewed Journals	Harper JD, Cunitz BW, Dunmire B, Lee FC, Sorensen MD, Hsi RS, Thiel J, Wessells H, Lingeman JE, Bailey MR. "First-in-human clinical trial of ultrasonic propulsion of kidney stones." J Urol. 2016 Apr;195(4 Pt 1):956-64. Epub 2015 Oct 30. http://dx.doi.org/10.1016/j.juro.2015.10.131 ; PubMed PMID: 26521719; PubMed Central PMCID: PMC4851928 , Apr-2016
Articles in Peer-reviewed Journals	Janssen KM, Brand TC, Cunitz BW, Wang YN, Simon JC, Starr F, Liggitt HD, Thiel J, Sorensen MD, Harper JD, Bailey MR, Dunmire B. "Safety and effectiveness of a longer focal beam and burst duration in ultrasonic propulsion for repositioning urinary stones and fragments." J Endourol. 2017 Aug;31(8):793-9. Epub 2017 Jun 26. https://doi.org/10.1089/end.2017.0167 ; PubMed PMID: 28537452 , Aug-2017
Awards	Maxwell A. "Promotion to Research Assist. Prof., March 2017." Mar-2017
Awards	Cunitz B. "Engineering and Urology Society Best Abstract award, May 2016." May-2016
Awards	Purington B. "University of Washington Distinguished Staff Award, July 2016." Jul-2016
Awards	Bailey M. "Promotion to Assoc. Prof., University of Washington, July 2016." Jul-2016
Awards	Harper J. "Promotion to Assoc. Prof., University of Washington, July 2016." Jul-2016
Awards	Lingeman J. "American Urological Association Distinguished Contribution Award, February 2017." Feb-2017
Awards	Simon J. "Promotion to Assist. Prof., January 2017." Jan-2017
Awards	Bruce M. "Best Poster. The 21st European Symposium on Ultrasound Contrast Imaging, June 2016. " Jun-2016
Books/Book Chapters	Lingeman JE, Beiko D, Bailey MR, Gettman MT, Kohrmann KU, Liatskos E, Matlaga BR, Monga M, Tailly G, Timoney A. "Stone Technology: Shock Wave and Intracorporeal Lithotripsy." in "Stone Disease. A Joint SIU-ICUD International Consultation Glasgow, Scotland, October 12–15, 2014." Ed. J. Denstedt, J. de la Rosette. Montreal: Société Internationale d'Urologie (SIU), 2015. p. 179-278., Jan-2015
Papers from Meeting Proceedings	Nikolaeva AV, Sapozhnikov OA, Bailey MR. "Acoustic radiation force of a quasi-Gaussian beam on an elastic sphere in a fluid." 2016 IEEE International Ultrasonics Symposium, Tours, France, September 18-21, 2016. In: 2016 IEEE International Ultrasonics Symposium. http://dx.doi.org/10.1109/ULTSYM.2016.7728608 , Sep-2016
Significant Media Coverage	Lyndon B. Johnson Space Center. "Article 'Soft-Tissue Emulsification Using a Mechanism of Ultrasonic Atomization Inside Gas or Vapor Cavities.' about PI Michael Bailey's work." NASA Tech Briefs, July 2016. MSC-25191-1. http://www.techbriefs.com/component/content/article/ntb/tech-briefs/bio-medical/24976 , Jul-2016
Significant Media Coverage	McGrane C. "Article about PI's research, 'NASA-funded UW researchers develop kidney-stone zapping technology.' " GeekWire July 4 2016. https://www.geekwire.com/2016/nasa-funded-uw-researchers-develop-kidney-stone-zapping-technology/ , Jul-2016
Significant Media Coverage	Bailey MR. "Dissolving Kidney Stones With Ultrasound." AIUM (American Institute of Ultrasound in Medicine) Soundwaves July 13, 2016., Jul-2016
Significant Media Coverage	NIH National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) research update. " 'Moving stones with sound—new ultrasound technology repositions kidney stones in people.' Article about PI's research and journal article in J Urol--Harper JD, Cunitz BW, Dunmire B, Lee FC, Sorensen MD, Hsi RS, Thiel J, Wessells H, Lingeman JE, and Bailey MR. First in human clinical trial of ultrasonic propulsion of kidney stones. J Urol 195: 956-964, 2016. " NIH NIDDK newsletter Jun 10 2016., Jun-2016
Significant Media Coverage	Focused Ultrasound Foundation. "News highlighting PI's research and new funding: 'Focused Ultrasound in Outer Space?' " Focused Ultrasound Foundation newsletter. July 21, 2016., Jul-2016
Significant Media Coverage	American Urological Association. "Newsletter references NIDDK article on 'Moving kidney stones with ultrasound.' referencing journal article in J Urol--Harper JD, Cunitz BW, Dunmire B, Lee FC, Sorensen MD, Hsi RS, Thiel J, Wessells H, Lingeman JE, and Bailey MR. First in human clinical trial of ultrasonic propulsion of kidney stones. J Urol 195: 956-964, 2016. " AUA newsletter Eureka June 2016., Jun-2016